

printcap — End-User Guide

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A friendly, step-by-step guide to using **printcap**, the network print-capture tool. This guide is written for the people who *operate* printcap day to day — no deep networking background required. Where something is genuinely technical, there's a plain-language explanation and a pointer to the Glossary.

For configuration-file reference, command-line flags, and deployment details, see the **ADMIN_GUIDE.md** instead. This document focuses on *using* the app.

Contents

1. What printcap does
 2. Before you start
 3. Installing & launching
 4. The 60-second Quick Start
 5. Capturing print jobs (the core feature)
 6. The web Dashboard
 7. The Settings tabs (Advanced view)
 8. Network packet capture (intercept mode)
 9. The Capture Window & reading packets
 10. Filtering like a pro
 11. Per-packet details
 12. Finding, saving & exporting your files
 13. Running as a Windows service
 14. Troubleshooting & FAQ
 15. Glossary
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1. What printcap does

printcap pretends to be a network printer. When a computer, server, or multifunction printer (MFP) sends a print job to printcap's IP address, printcap **accepts the job and saves it** — both the raw data and a readable copy of the document where possible — instead of putting ink on paper. It speaks every common print "language" (RAW/JetDirect on port 9100, LPR/LPD on 515, IPP on 631, and more), answers printer-discovery requests (SNMP, Bonjour/AirPrint, WSD) so it looks like a real device, and shows everything it catches in a live web dashboard.

It has two big jobs:

- **Print capture** — collect and inspect the documents sent *to* it. This is the everyday use and needs no special privileges.

- **Network capture (intercept mode)** — record raw network traffic on an adapter so you can analyze what a printer and its clients are actually saying to each other. This is an **authorized-use-only** diagnostic/security feature (see Section 8).

You can drive printcap three ways, and they all share the same settings:

Surface	Best for
Windows app (GUI)	Day-to-day operation on a desktop. Has a Quick Start tab for newcomers and Advanced tabs for everything else.
Web Dashboard	Watching captures live, from any browser, including on another machine.
Command line / service	Unattended/background operation (see Section 13).

2. Before you start

- **A Windows PC** (printcap is Windows-first; it also runs on macOS/Linux for capture). You usually need to run it **as Administrator** so it can open low-numbered ports (like 515 and 631) and the firewall.
- **The printcap program file** (printcap.exe). There are two builds:
 - printcap.exe — the **full** build, includes live network packet capture (needs **Npcap**, see below).
 - printcap-nocapture.exe — print capture only, no packet-capture engine.
- **Npcap** (*only if you'll use network packet capture*) — a free Windows component that lets programs read network traffic. printcap will detect if it's missing and offer to take you to the download page. See "PCAP / Npcap" in Troubleshooting.
- **Authorization** (*only for intercept mode*) — written permission to capture traffic on the network you're testing. printcap **will not start** intercept mode until you fill in who you are and your engagement reference.

3. Installing & launching

printcap is a **single self-contained file** — there's no installer for the app itself. To run it:

1. Copy printcap.exe to a folder you can write to (e.g. C:\printcap\).
2. **Right-click** → **Run as administrator**. The main window opens.
3. The first time, printcap creates a settings file (printcap.json) and a capture folder next to the program.

If Windows SmartScreen warns about an unrecognized app, choose **More info** → **Run anyway** (the file is unsigned). Your security team can verify the build hash shown by printcap.exe - version.

4. The 60-second Quick Start

The **Quick Start** tab (the first tab, designed for non-technical staff) is all most people need:

1. Open printcap. You land on **Quick Start**.
2. Read the **Status** line — it shows whether capture is running.
3. Click **Start**. printcap is now listening as a printer.
4. On another computer, **add a printer** pointing at this PC's IP address (any common type works — see Section 5) and print a test page.
5. Click **Open Dashboard** to watch the job arrive, or **Open Capture Folder** to see the saved files.
6. Click **Stop** when you're done.

Other one-click buttons on Quick Start:

- **Open Capture Window** ▶ — watch live network packets (intercept mode).
- **Save Settings** — write your configuration to disk.
- **Help** — the built-in, plain-language help for every feature.

Everything beyond this is the **Advanced view** — the other tabs across the top (Section 7). Quick Start never hides anything; it's just the simple front door.

5. Capturing print jobs (the core feature)

5.1 Turn on the protocols you need

On the **Protocols & Ports** tab, switch on the print methods you expect and set their ports. The common ones:

Protocol	Port	Used by
RAW / JetDirect / AppSocket	9100	Most network printers and Windows "Standard TCP/IP" ports
LPR / LPD	515	Enterprise systems — SAP, IBM AS/400, z/OS mainframe, Linux/CUPS
IPP / IPPS	631	Modern printing, AirPrint, CUPS

Set further options on the **RAW & LPR** and **IPP & TLS** tabs (resource paths, TLS certificate, per-queue mainframe settings, etc.).

5.2 Choose where and how jobs are saved

On the **General** tab:

- **Capture directory** — where saved jobs and logs go. (Logs land in a logs sub-folder.)
- **Save mode:**
 - ▶ **Both** (*default*) — save the raw spool data **and** a small .json description of each job.
 - ▶ **Raw** — just the document bytes.

- **Meta** — just the description; the document itself is discarded.
- **Max job size** — a safety cap so a runaway job can't fill your disk (0 = no limit).

5.3 Make this PC discoverable (optional)

On the **Discovery** tab you can answer **mDNS/DNS-SD** (Bonjour/AirPrint) and **WSD** so the fake printer shows up in "Add Printer" browsers automatically. On the **SNMP** tab, printcap can answer printer-discovery queries (including secure **SNMPv3**) so it looks like a real device to management tools.

5.4 Send a test job and confirm

Print anything to this PC's IP from another machine. Within a second or two the job appears under **Captured jobs** in the dashboard, and the files appear in your capture folder. printcap automatically detects the document type (PDF, PostScript, PCL, JPEG, plain text, EBCDIC mainframe data, ...) and names the readable copy accordingly.

6. The web Dashboard

Click **Open Dashboard** (or browse to `http://<this-pc-ip>:<dashboard-port>/`). The dashboard has these sections:

- **Settings** — the same configuration as the GUI tabs, editable from a browser. Secrets (passwords, community strings) are shown as *** and preserved unless you change them. *Changes from another machine are read-only unless an admin has enabled remote administration.*
- **Network capture** — the live/saved packet viewer (Sections 9–11).
- **Listeners** — which print protocols are up, and buttons to **Start** / **Restart** / **Stop** the engine.
- **Captured jobs** — every job caught, searchable, with **CSV/JSON export** and per-job preview/download.
- **Live log** — a real-time, filterable log (error → trace) with a one-click **download full log**.

There's a **light/dark theme** toggle, and the dashboard auto-refreshes.

7. The Settings tabs (Advanced view)

Every feature is configurable through the GUI tabs. Each tab has a matching page in the in-app **Help** window.

Tab	What it controls
Quick Start	The simple landing page (status + big action buttons).
General	Capture folder, save mode, job size cap, notifications.
Protocols & Ports	Which print protocols/ports are enabled.
RAW & LPR	RAW/9100 options; LPR/LPD queues and per-queue mainframe defaults.
IPP & TLS	IPP/IPPS resource paths and the TLS certificate.
Printer Identity	The make/model/name printcap reports as.
SNMP	SNMP v1/v2c identity and secure SNMPv3 (USM) users.
Discovery	mDNS/DNS-SD (Bonjour/AirPrint) and WSD advertisement.
SMB Share	Experimental SMB2/3 print share.
Mainframe / EBCDIC	Decode mainframe (EBCDIC) print streams to readable text.
Forward Proxy	Relay captured jobs onward to a real printer.
Content / DLP	Scan captured documents for sensitive content and flag matches.
Logging	Log file, verbosity, and SIEM export (JSON-lines + syslog).
Capture	Network packet capture / intercept mode (Section 8).
Service & Firewall	Install as a Windows service; open firewall ports.

8. Network packet capture (intercept mode)

Intercept mode records raw network traffic on a chosen network adapter into a **.pcap** file (the standard format Wireshark reads) and shows it live. Use it to see exactly what a printer and its clients exchange.

⚠ Authorized use only. Capturing network traffic may only be done on networks you have **written permission** to test. printcap enforces this: on the **Capture** tab, the **Authorization** box (your name + engagement/ticket reference, with an optional expiry) **must be filled in or capture refuses to start**. Those details are stamped into a record saved next to every capture.

8.1 The simplest case — passive capture

1. On the **Capture** tab (or in the Capture Window), pick the **Printer network adapter** — the network card connected to the printer's network.
2. Fill in the **Authorization** box.
3. Click **Start capture**. Packets stream into the viewer and into a **.pcap** file in your capture folder.

This is **passive** — printcap just listens. It does not change the network.

8.2 Focus on one printer (MFP)

Set the **MFP / printer IP** field to the printer's address. Two things happen:

- The "**MFP only**" toggle in the Capture Window filters the view to just that printer's traffic.
- (Windows) if you turn on active positioning, that IP is automatically the in-scope target.

8.3 Keeping a printer online while you capture (multi-homed + NAT)

Windows only, advanced. If you place the PC **between** a printer and the rest of the network (two network adapters), printcap can route the printer's traffic so it keeps working while you watch it:

- **Printer network adapter** — the side the printer is on.
- **Internet adapter (uplink)** — the side with internet/network access.
- **Auto-NAT (ICS)** — fill in the two connection-name boxes (e.g. Ethernet and Wi-Fi, chosen from the drop-downs) and printcap turns on Windows Internet Connection Sharing when you Start and **turns it off again when you Stop**.

Note: ICS gives the printer side the fixed address 192.168.137.1. It fits the "this PC is the printer's router" setup. For capturing on an *existing* network where the printer keeps its own address, your admin should use RRAS NAT instead. Leave the ICS boxes blank to skip auto-NAT.

8.4 Cleanup is automatic

When you click **Stop** (or close the app), printcap puts everything back the way it found it — any network changes it made are reverted, and Internet Connection Sharing is switched off. The Capture Window confirms this in its status line.

8.5 Reconstruct documents from captured traffic (carve)

On the **Capture** tab, **Reconstruct files (carve)** rebuilds whole print jobs out of the captured packets and saves them as typed files (.pdf, .pcl, .ps, .jpg, ...) alongside the raw .pcap — so you get the documents, not just the packets.

9. The Capture Window & reading packets

Open it with **Open Capture Window** ► (Quick Start) or from the dashboard's **Network capture** section.

9.1 Live vs. saved

- **Live** — a scrolling, real-time view as packets arrive. It shows a "missed N" notice if traffic outruns the buffer.
- **Static / Refresh** — reads the saved .pcap from disk, with paging.

9.2 The columns

Column	Meaning
#	Packet number (click it on the web to open full details).
Time	When it was captured.
Proto	TCP, UDP, ICMP, ARP , or non-IP, plus a service tag (HTTP, IPP, raw...).
Source / Destination	Who sent it / who it's going to. Click an IP to filter to it.
Len	Size in bytes.
Info	A human-readable summary (ports, TCP flags, ARP "who-has", ICMP type...).

9.3 Row colors

- **Red** — errors and connection resets (problems worth noticing).
- **Green** — print jobs (RAW/9100, LPR, IPP) and SNMP.
- **Blue** — encrypted HTTPS (port 443).

9.4 Row types you'll see

- **TCP / UDP / ICMP** — normal traffic, fully decoded.
- **ARP** — "who has 10.0.0.9? tell 10.0.0.1" / "10.0.0.9 is at ". You'll see a lot of these during active positioning; that's expected.
- **non-IP** — other low-level frames; the Info column shows the EtherType.

9.5 Following a conversation

Double-click any TCP row to **follow the stream** — printcap reassembles both directions (client→server and server→client). Web/API traffic (HTTP) is shown as readable text, so printer web pages, EWS, REST, and IPP exchanges are legible. Each direction is downloadable, and the whole .pcap is one click away.

10. Filtering like a pro

The viewer has a **Wireshark-style display filter**. Type terms in the **Filter** box; separate terms with spaces (they all must match), and a plain word matches anywhere in the row.

10.1 Fields and operators

Group	Fields	Operators
Addresses	src dst addr ip	== != ~ (contains)
Ports	sport dport port	== != > < >= <=
Classification	proto svc class color info	== != ~
Numbers	len ipver	== != > < >= <=

10.2 Examples

Type this	To see
<code>addr==10.0.0.50</code>	only the printer at 10.0.0.50
<code>port==9100</code>	print (RAW/JetDirect) traffic
<code>svc==http</code>	the printer's web pages / API
<code>proto==arp</code>	ARP "who-has / is-at" messages
<code>class==reset</code>	dropped/reset connections
<code>ipver!=6</code>	hide IPv6
<code>dst==10.0.0.50 len>200</code>	big packets going to the printer

10.3 Shortcuts

- **Quick filters** — one-click buttons (Print 9100, IPP, Web/API, ARP, Resets, ICMP errors, SNMP) fill the filter box for you.
- **Click any IP** in a row to instantly filter to that address.
- **Click a column header** to sort by it (click again to reverse).
- **Hide IPv6** checkbox — the same as typing `ipver!=6`.
- **MFP only** checkbox — limits the view to the printer IP you set.

10.4 Capture-time filter (keep the file small)

The filters above narrow what you *see*. To control what gets *recorded*, set **intercept.capture_filter** (same syntax) — only matching packets are written to the `.pcap`. Handy for long captures. (On Windows, `intercept.bpf` is an alternative using classic libpcap syntax; it's Windows/Npcap-only.)

11. Per-packet details

To see everything about one packet:

- **Web dashboard:** click the packet's # number.
- **Windows Capture Window:** click a row to select it, then click **Packet details**.

A popup shows the **layer-by-layer decode** — Ethernet (MAC addresses, EtherType), IP (addresses, TTL, protocol), TCP/UDP (ports, flags, sequence numbers), ICMP, or ARP — followed by a **hex + ASCII dump** of the raw bytes, just like Wireshark's detail pane. If the packet was only partially captured, the dump says so and points you to the full `.pcap`.

Tip: in the live Windows view, your selected row stays selected even as new packets arrive, so you can open details mid-capture without the row jumping away.

12. Finding, saving & exporting your files

Everything lands under your **Capture directory** (set on **General**):

- **Captured print jobs** — raw spool files plus .json descriptions, and any reconstructed documents (.pdf, .pcl, ...).
- **Network captures** — capture.pcap (open in Wireshark) plus a small record of who ran the capture and under what authorization.
- **Logs** — in the logs sub-folder; download the current log from the dashboard's **Live log** section.

From the dashboard you can **export the job list as CSV or JSON**, preview or download any single job, and download the full .pcap.

13. Running as a Windows service

To capture around the clock without anyone logged in, install printcap as a Windows **service** from the **Service & Firewall** tab (or the command line). The service starts at boot and runs unattended. The same tab can open the required ports in **Windows Defender Firewall**.

Live capture & the service: when printcap runs as a service, packet capture happens inside the *service* process. The desktop Capture Window can't show those packets live — use the **web dashboard's** Network capture view on the service machine instead, or stop the service and capture from the GUI.

14. Troubleshooting & FAQ

Npcap / "PCAP" not installed

Live **network** capture needs **Npcap** (the Windows packet-capture driver). If it's missing, printcap tells you and offers the download page (<https://npcap.com>). Install it (default options are fine), then restart printcap. You also need the **full** printcap.exe build — the printcap-nocapture.exe build has no capture engine. (*Print-job capture does **not** need Npcap.*)

"Port already in use" / a listener failed

Another program (often the Windows Print Spooler or a real printer driver) is already using the port. printcap names the failing listener and the reason on the **Listeners** panel and in the log. Free the port, change printcap's port for that protocol, or stop the conflicting program.

Nothing gets captured when I print

- Is the engine **Started**? Check the Status line / Listeners panel.
- Did you point the other computer at **this PC's IP** and the **right port** (9100 for most)?
- Is **Windows Firewall** blocking the port? Use **Service & Firewall** to open it.
- Are you running printcap as **Administrator** (needed for ports below 1024 like 515/631)?

I see IPv6 packets I don't want

Use the **Hide IPv6** checkbox (or filter `ipver!=6`) to hide them from the view immediately. To stop recording them entirely, tick **Disable IPv6** before you **Start** capture (it takes effect on the next start).

"ring overrun — older packets dropped"

Traffic arrived faster than the live view could keep up, so the oldest live packets scrolled out of memory. The **.pcap file on disk is still complete** — open it with **Refresh (static)** or in Wireshark to see everything.

I can't select a packet row in the Capture Window

This was fixed: single-click selects a row and the selection now sticks during live capture. If you're on an older build, update to the latest. Then select a row and click **Packet details**, or double-click to follow the stream.

What are all the "ARP" / "non-IP" rows?

ARP is the normal "who-has / is-at" address chatter on a network — you'll see lots during active positioning. "non-IP" rows are other low-level frames; the Info column shows what kind.

My settings didn't save

Click **Save Settings** (Quick Start) or **Save** in the GUI. From the web dashboard, settings changes are **read-only from other machines** unless an admin has enabled remote administration.

15. Glossary

Term	Plain meaning
MFP	Multifunction printer (print/scan/copy device).
RAW / 9100 / JetDirect	The most common network print method (TCP port 9100).
LPR / LPD (515)	An older/enterprise print protocol (SAP, AS/400, mainframe, Linux).
IPP / IPPS (631)	Modern internet printing; IPPS is the encrypted version.
SNMP	How management tools discover and query devices; SNMPv3 is the secure version.
pcap / .pcap	The standard capture-file format that Wireshark reads.
Npcap	The free Windows component that lets programs read network traffic.
Packet	One small chunk of data sent across the network.
ARP	The network's "who has this address?" address-lookup chatter.
MAC address	A network card's hardware address (e.g. a4:5e:60:...).
TCP / UDP / ICMP	Common network transport types; ICMP includes "errors"/ping.
TTL	"Time to live" — how many hops a packet may travel before being dropped.
NAT / ICS	Address translation that lets one network reach another; ICS is Windows' built-in version.
Intercept / on-path capture	Recording traffic by placing the capture host in its path. Authorized use only.
Carve	Rebuilding whole files/documents out of captured packets.
DLP	Data-loss prevention — scanning documents for sensitive content.
EBCDIC	The text encoding used by IBM mainframes.

printcap — authorized print and network-capture use only. When in doubt about permission to capture on a network, stop and ask.